

Sinusoidal Graphs: Amplitude

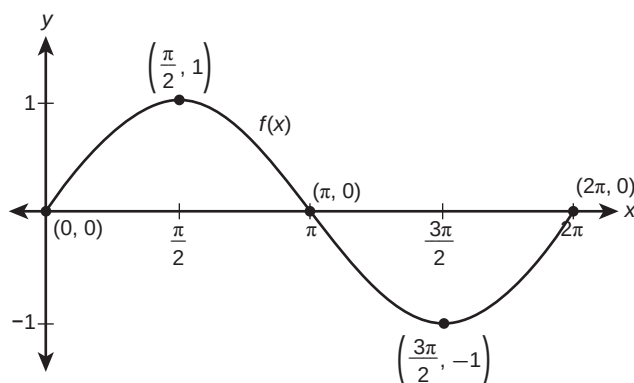
The amplitude is the height of a sinusoidal function above the midline.

Defining the Sine Function

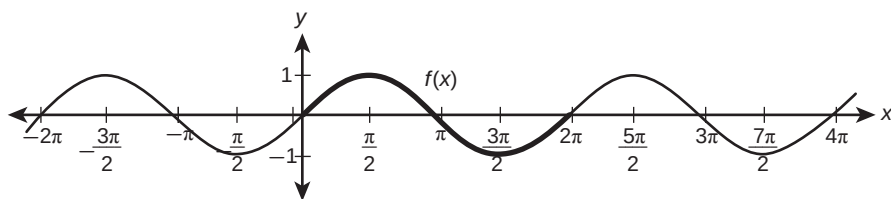
Recall some key points from the unit circle for sine.

Angle (in radians)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
Sine ratio	0	1	0	-1	0

Now consider the angles as inputs (x -values) and the sine ratios as outputs (y -values). Plot those points, and draw a smooth curve through them. You will see that the result is a sinusoidal function, $f(x) = \sin x$.



Since you can find the sine of any angle measure, the graph extends horizontally in both directions without end. Therefore, the domain of $f(x) = \sin x$ is $(-\infty, \infty)$.



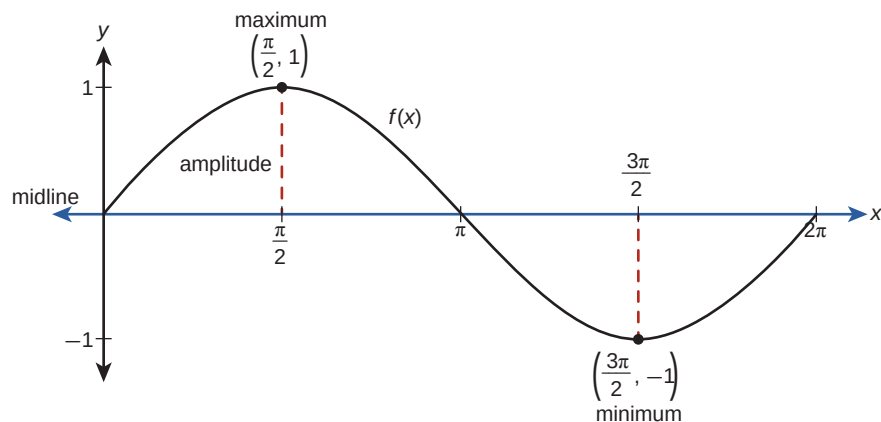
► **Think About It** Since the graph of $f(x) = \sin x$ is symmetric about the origin, it is an odd function $f(-x) = -f(x)$.

Since the graph repeats every 2π , the period of $f(x) = \sin x$ is 2π . Also, since $\sin(0) = 0$, the graph of $f(x) = \sin x$ passes through the origin $(0, 0)$.

The x -intercepts occur whenever sine is equal to zero ($x = \dots -2\pi, -\pi, 0, \pi, 2\pi, \dots$), so the x -intercepts occur at $x = n\pi$ where n is an integer.

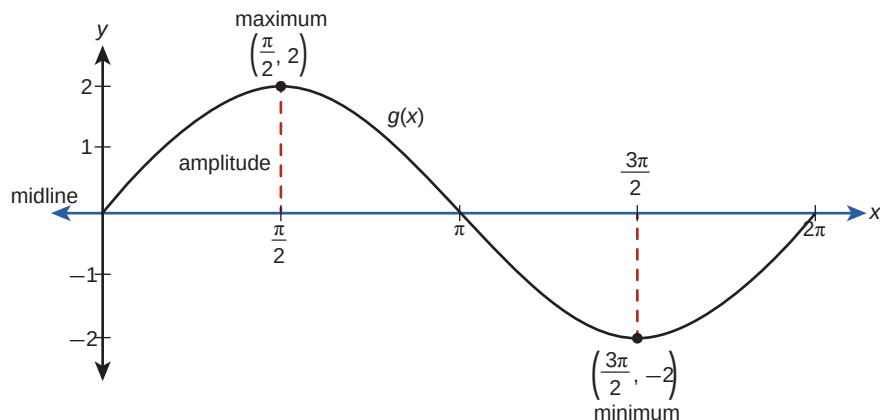
Inspecting the graph reveals that a maximum is $\left(\frac{\pi}{2}, 1\right)$ and a minimum is $\left(\frac{3\pi}{2}, -1\right)$. Therefore, the range of $f(x) = \sin x$ is $[-1, 1]$.

The midline is halfway between the minimum y -value ($y = -1$) and the maximum y -value ($y = 1$), so the equation of the midline is $y = 0$. Since the amplitude is the height above the midline, the amplitude is equal to 1.



EXAMPLE 1

Determine the amplitude of $g(x) = 2 \sin x$, which is graphed.



SOLUTION

Notice that $g(x) = 2 \sin x$ has the same basic shape as $f(x) = \sin x$ but is stretched vertically by a factor of 2. Since the maximum is 2 and the minimum is -2 , the amplitude of $g(x) = 2 \sin x$ is 2. ■

Formula for the Amplitude of a Sinusoidal Function

The amplitude of a sinusoidal function can be found directly from the function equation. For $f(x) = A \sin x$,

$$\text{amplitude} = |A|$$

where $A \neq 0$.

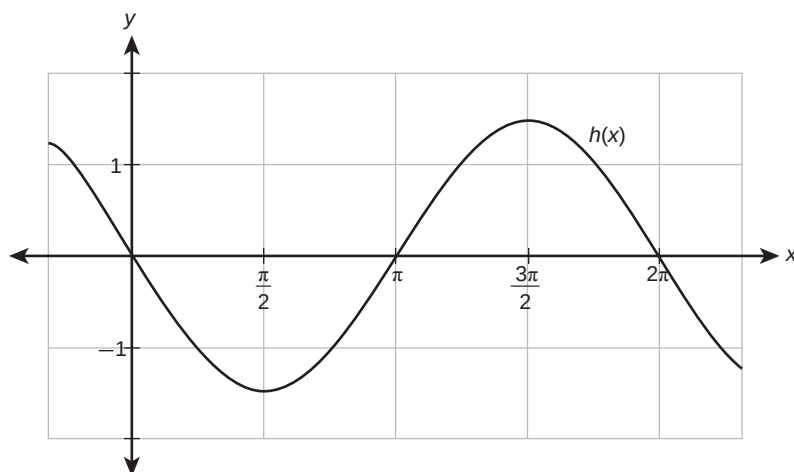
EXAMPLE 2

Determine the amplitude of $h(x) = -\frac{3}{2} \sin x$.

SOLUTION

Since the function is in the form $h(x) = A \sin x$ and $A = -\frac{3}{2}$, the amplitude is $\left| -\frac{3}{2} \right| = \frac{3}{2}$.

Confirm this result by graphing it.



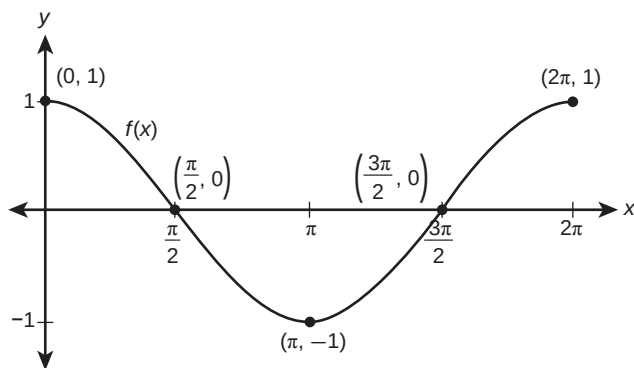
Notice that compared to the parent function $f(x) = \sin x$, the graph of $h(x) = -\frac{3}{2} \sin x$ is stretched vertically and is also reflected over the x -axis, which occurs because $A < 0$. The graph of $h(x) = A \sin x$ is the result of reflecting the function $y = |A| \sin x$ over the x -axis if $A < 0$.

Defining the Cosine Function

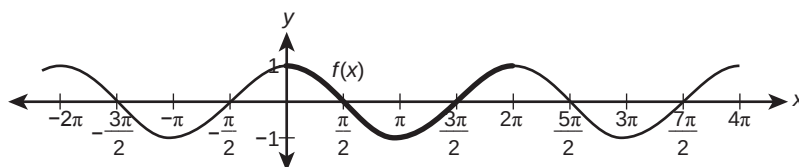
Recall some key points from the unit circle for cosine.

Angle (in radians)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
Cosine ratio	1	0	-1	0	1

When graphed using a smooth curve (where x = angle measures, y = cosine ratios), you get the graph of the cosine function, $f(x) = \cos x$.

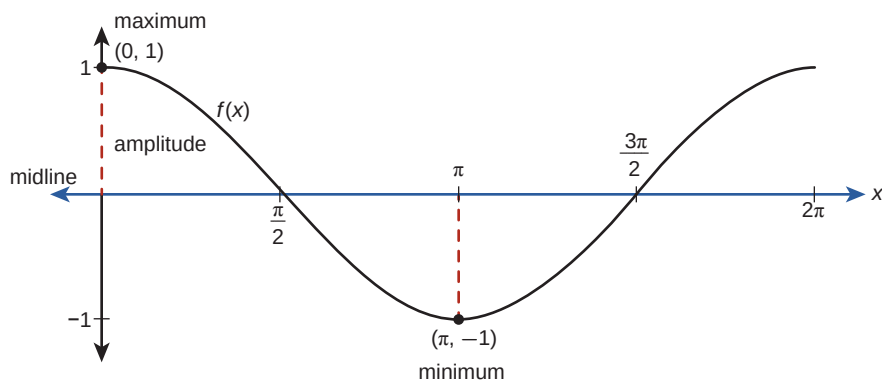


Just like the sine function, this pattern repeats and extends horizontally in both directions, so the domain of $f(x) = \cos x$ is $(-\infty, \infty)$, and the period is 2π . Notice that the range of $f(x) = \cos x$ is also $[-1, 1]$, as with the sine function.



The midline is also at $y = 0$, and the amplitude is 1.

► **Think About It** Since the graph of $f(x) = \cos x$ is symmetric about the y -axis, it has even symmetry $f(-x) = f(x)$.

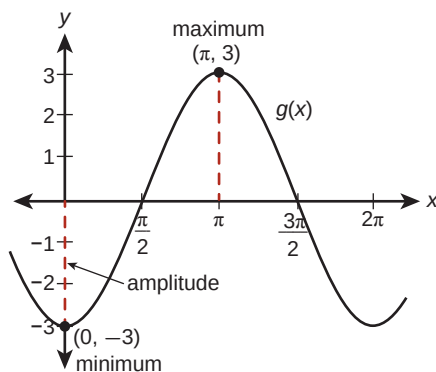


However, note the following differences for the cosine function:

- The y -intercept is $(0, 1)$.
- A maximum occurs at $(0, 1)$, and a minimum is at $(\pi, -1)$.
- The x -intercepts occur where the cosine function is equal to zero, which are the odd multiples of $\frac{\pi}{2}$: $x = \left\{ \dots -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots \right\}$. So the x -intercepts occur when $x = \frac{n\pi}{2}$, where n is an odd integer.

EXAMPLE 3

Determine the amplitude and the equation in the form $g(x) = A \cos x$.



► Q&A

Q What is the range shown in Example 3?

A $[-3, 3]$

SOLUTION

$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2} = \frac{3 - (-3)}{2} = 3$$

Since the graph is reflected over the x -axis, $A < 0$ in the form $f(x) = A \cos x$.

So the equation of the function is $g(x) = -3 \cos x$. ■

EXAMPLE 4

Determine the amplitude of $f(x) = 3.5 \cos x$.

SOLUTION

Since the function is in the form $f(x) = A \cos x$ and $A = 3.5$, the amplitude is $|3.5| = 3.5$. ■